

Table Tennis Pro: Smart Training for Global Enthusiasts

Brain-inspired architecture-
based mobile intelligent agent

SMART RELEASE SPORTS FUN

Track

Main track of higher education

Group

Undergraduate - Creative Group

Invited institutions

Beijing University of Posts and Telecommunications

School Group

Queen Mary University of London The University of Texas at Austin
University of Applied Sciences Technikum Wien



Market prospect

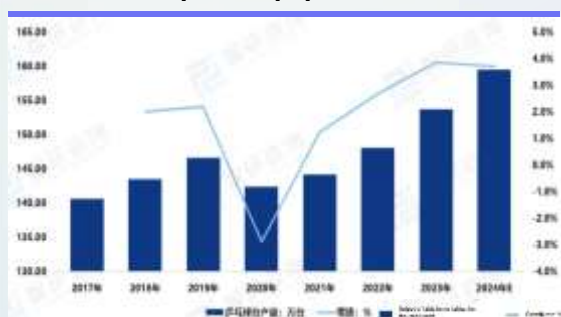
Opportunities and challenges coexist in the global sports industry

The total value of the sports industry has been rising steadily, and the table tennis industry has entered a golden period of development

- According to the historical development experience of many countries, the sports industry grows quickly once per capita GDP hits 10,000 US dollars.
- The scale of China's table tennis industry is expected to reach 70 billion yuan by 2025, and has become an important engine driving the overall consumption growth.
- The overseas sports market (narrow market) is expected to reach 4.5 trillion yuan (\$623 billion) by 2027.

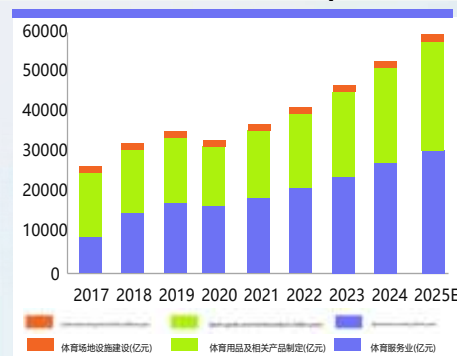


Global table tennis market growth (sports equipment)



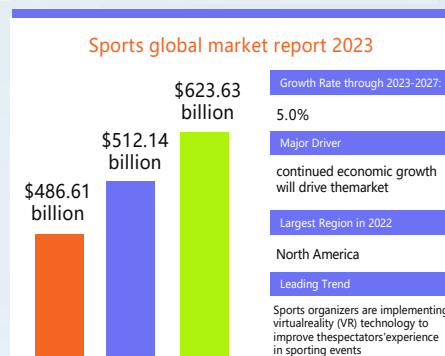
Data source: eastmoney.com

China Sports Industry Output Forecast (calculated based on output value)



Data source: China National Bureau of Statistics, Report of the State Council's "Comprehensive Fitness Plan"

Overseas Sports Market Size Report (Narrowly Defined Market)



Data source: The Business Research Company

The global sports industry is about to take off

Market prospect

Opportunities and challenges coexist in the global sports industry

The problem of sports industry development needs to be solved urgently

Sports
training

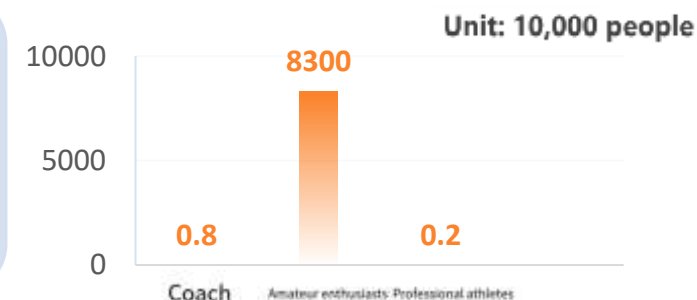
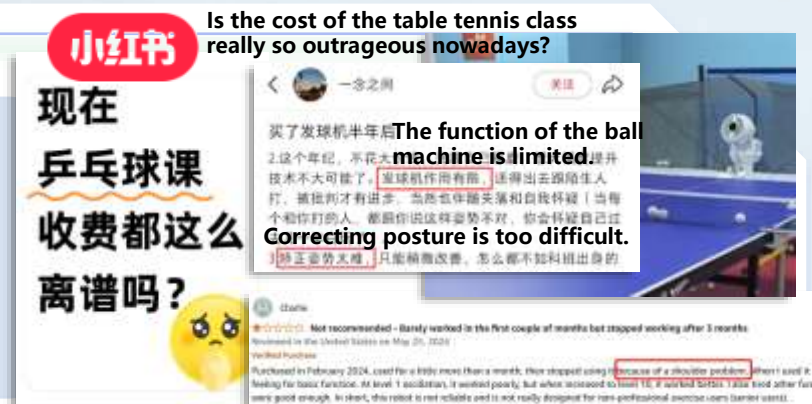


- ◆ Extremely scarce and uneven professional coaching resources
- ◆ Most people can't afford the expensive training
- ◆ The tedious process of self-improvement

Sports
competitions



- ◆ Amateurs generally lack the means of technical and tactical analysis and review
- ◆ Quality sports resources are difficult to sink, and amateurs lack coaching training and competition opportunities



In the next five years

The number of professional and standardized table tennis events in the private sector increased by about **250%**

The technical analysis and review function was used by **40%** of the contestants

AI coaches make up about **60 %** of the amateur coaching gap

- »» The **intelligent and zero-threshold** table tennis magic mirror system advocates **green events** and contributes to the **construction of ecological civilization**.
- »» By addressing relevant employment, social and environmental issues, we will improve **the quality of life and happiness of the people**, demonstrating our commitment to **improving peoples well-being**.

The inspection process

1

小红书

以后去球馆
练球，发球
机不敢用了

From now on, when
I go to the gym to
practice, I won't use
the serve machine
anymore.

The problem was exposed when
the players complained

3

发球机卡顿中。。。它怎么时不时就卡顿

The ball machine is
malfunctioning...
Why does it sometimes
get stuck and freeze?

User interview, in-depth
communication

5

全球乒乓球产业增长 (体育器械)



Industry reports
market analysis



2



Watch the gym and record
it on the spot

4



Enterprise visit, functional
investigation

6

Low rating,
low sales volume

美团



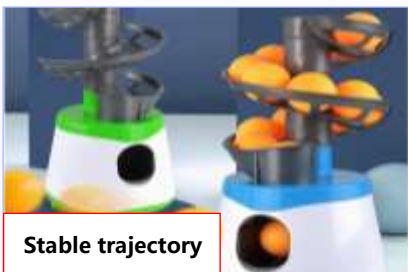
Platform research and
data analysis

Three core pain points



The training was not targeted

"I have been practicing for six months and I still can not handle side spin."
—from more than 300 amateur players



After-sales feedback of JD.com and Taobao APPs



Lack of post-match review

"I trained several times, but I still could not find the pattern of backhand errors."
—After-sales feedback of the server

小红书

Worry-free (Official Verification)

Want to ask everyone, for a beginner in table tennis, is it recommended to play against a ball machine?

Hitting against a ball machine is meaningless, the ball is different from the one played by a person, so it is not recommended to play, it will make people stupid, if you want to practice, you must have someone watching and guiding (the reason is very simple, you may not be able to see your own action problems), correct practice process is to find someone to hit multiple balls, to fix the action at the beginning, to practice the steps at the intermediate level, from fixed points to left and right, and then front and back steps, and then multiple balls with single balls, gradually transitioning to actual combat, the description is relatively simple, and the system needs time to be.

The proportion of amateur event service types

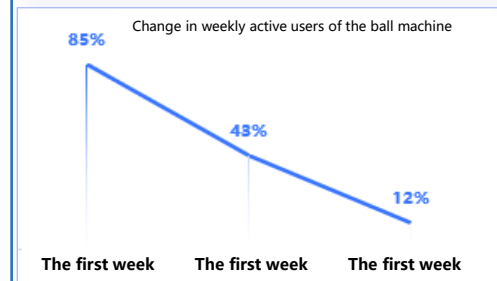


Little Red Book, "Table Tennis World" 2024 Amateur Tournament Research Report



Lack of incentive feedback

"I don't even know if I'm doing it right, and I get bored after a few times."
— Feedback from players



Zhihu, JD Consumer Research Institute "2024 Sports Intelligent Equipment Usage Report"

Pain point effect diagram

- ✗ No targeted training
- ✗ No review feedback
- ✗ No incentive mechanism



The overall training effect is poor



User satisfaction declines



The willingness to buy back is weak



Industry value

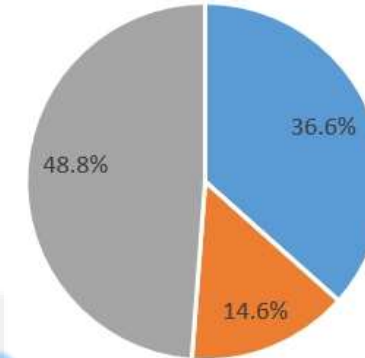
AI coaches empower and release 300 million market space

Brand	Brand image	merit	shortcoming
PongBot		High cost performance; strong structure and convenient installation	It does not support ball recovery; the rotation direction needs to be manually adjusted; there is no replay function; there is no intelligent feedback
Butterfly Amicus Prime		Three wheel design supports rich rotation	The package is expensive; the pitch is unstable; there is no replay function; there is no intelligent feedback
HP		Portable and user-friendly	There is no review function; there is no intelligent feedback system; the precision of movement is limited
DHS		The hardware is stable and widely used in domestic training sites	The function is single, without review function; lack of intelligent feedback and training plan support



Its equipped with "Ping Xiaozhi"

A total of about **\$123 million** in annual revenue was added



Revitalize the stock

Revenue increased by **45 million**
Wake up **50,000** ball machines that are eating dust

Incremental activation

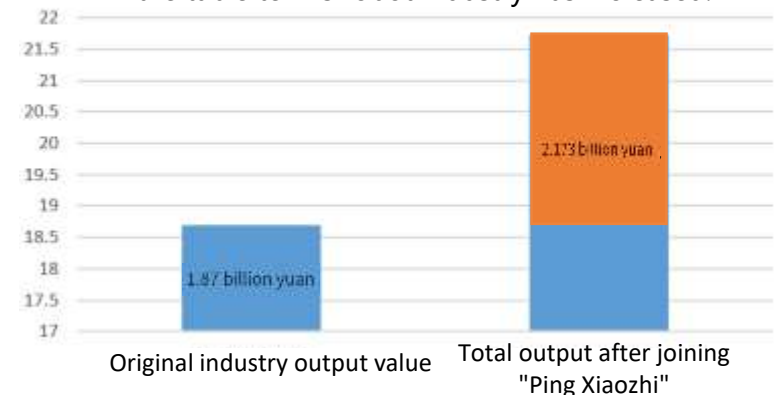
Revenue increased by **18 million**
Empower the sales of serving machines, exceeding **190 million**

Constant conversion

Revenue increased by **60 million**
Reach about **20%** of active users

The total output value of the industry increased by **303 million yuan**, with an increase of **16%**

After the addition of Ping Xiaozhi, the output value of the table tennis robot industry has increased.



Our answer

Launch a global table tennis training platform based on artificial intelligence



Ping Xiao Zhi

The first intelligent sports in China Training platform



Forehand Three-Point Forehand Skill

- Training Count: 20 balls
- Key Technical Notes
- 1. Insufficient forearm contraction
- 2. Body leaning forward not enough
- 3. Contact point too late
- Speed Range: 14.1 m/s - 14.5 m/s



Target pain points and fill the gap

Pain spot

Solution

① No teaching ability

The server has no teaching and correction ability.
Users blindly practice to deepen the wrong muscle memory.

① Intelligent teaching

Intelligent coach & data collection

Built-in intelligent coach, identify and record users actions and hitting effect, and guide users to correct in real time.

② Single mode

The server has a single mode of serving.
Its out of touch with reality.

② Multiple modes

Intelligent connection hub

The intelligent APP controls the ball delivery of the server, and can flexibly set the serving parameters, rotation and landing point combinations.

③ No training plan

The server does not provide a training plan.
Users practice without direction.

③ Intelligent solutions

Intelligent agent systems

The built-in MM-LLM personalized training program can improve the sports ability.

Scene display

Personalized and home training scenarios — Home training, no complicated wiring

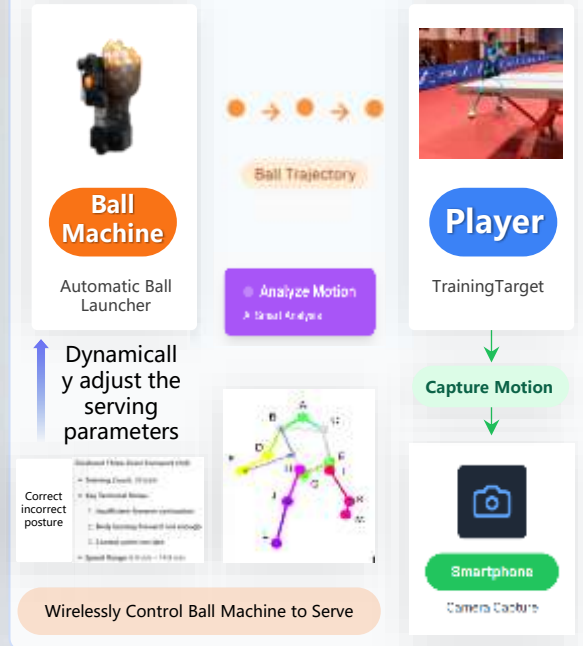


Open the Ping Xiao Zhi APP
View personalized daily tasks

Ball throwing
practice

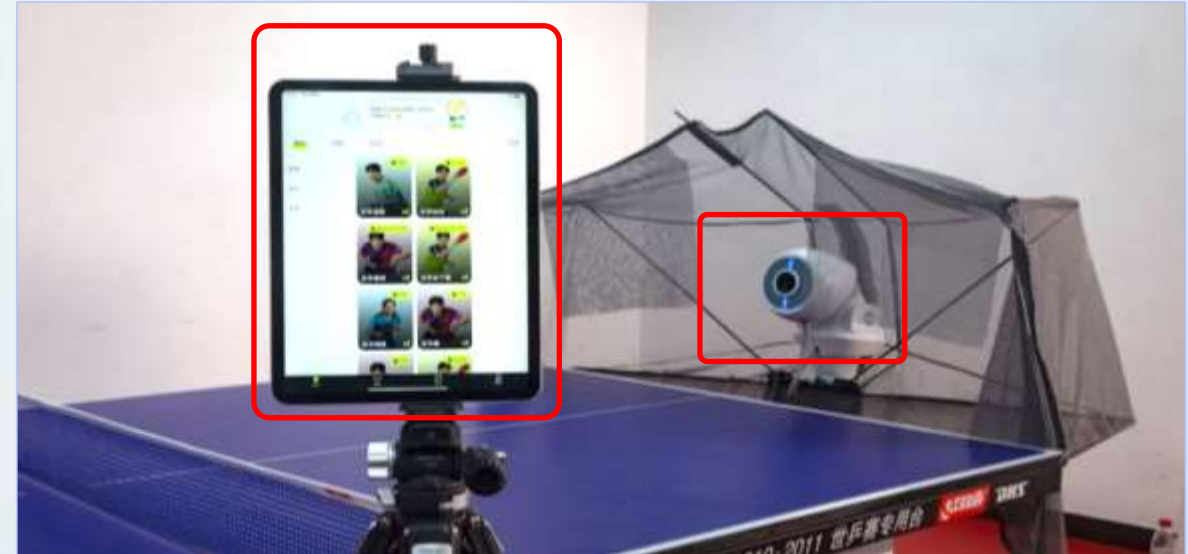


Connect the server with one
button of Bluetooth



Set the "ball pull practice" on
voice or touch screen
The server responds to your
personalized needs immediately

Selecting the appropriate
training program



Scene display

Multi-scenario application — No complex wiring is required

Home, school, gym.....All venues are one click away

The ball is served by the server through wireless control

User performance

Promote
Ball parameters, speed, spin, arc Angle...
Adjust
Help users step by step

PDA
One-click connection

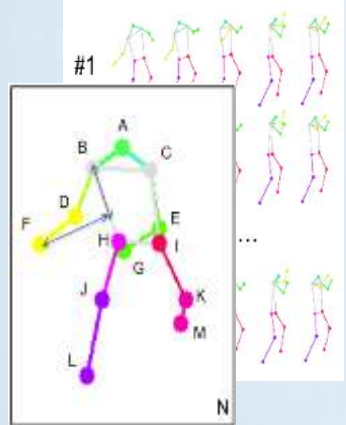
connected to the serving machine
Equipment is ready.

Voice animation guidance

Identify the motion of the sphere

Capture and split the action

- Core trunk
- Arms, wrists and fingers
- Page orientation
- hitting point.....



Action Recognition & Segmentation Method Based on Graph Neural Network

An intelligent tactical and technical analysis method covering the entire process

Recognizing and reconstructing the trajectory of table tennis based on tree pruning

Automatic optimization algorithm for Ball Machine
Full-process intelligence of Brain-like Intelligent Agent

Constructing an intelligent agent architecture resembling the human brain (industry's first) to achieve intelligent analysis



Automatic optimization algorithm for the ball machine



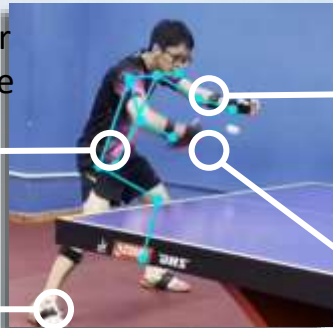
Key Entity Recognition
Temporal spatial correlation
Event segmentation
Semantic element extraction

Shift the center of gravity to the waist

footwork

Arm position

The layout is oriented towards the trajectory of the racket swing



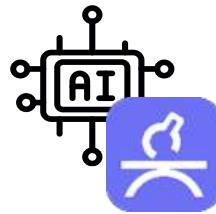
Application scenario value decomposition

Intelligent Connection Hub
& Intelligent Coach

Smart Upgrade Package



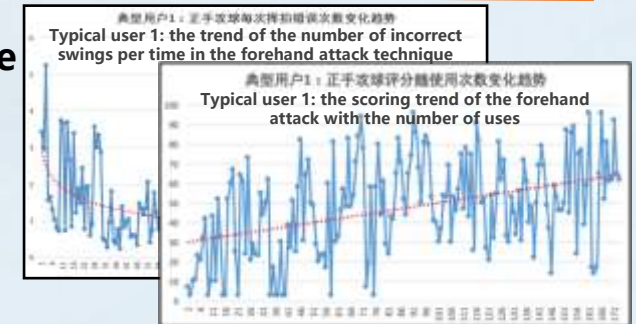
Traditional Ball Machines



Ping Xiaozhi
Serve Machine
Version

No replacement of the traditional ball machine, only an "intelligent upgrade package"
Compatible with interfaces of multiple brands of traditional ball machines,
flexible adjustment of serving parameters

The player uses Wenzi Zhi for specialized practice of forehand pull shots.
The system automatically collects the trajectory, spin direction and speed data of each shot.
This helps to clearly review the training process and accurately identify the problems.

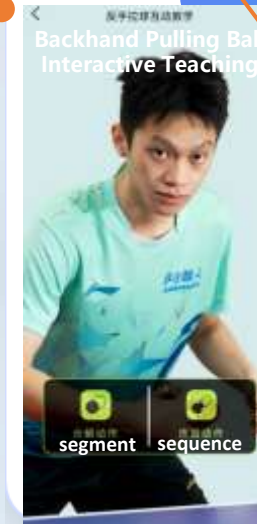
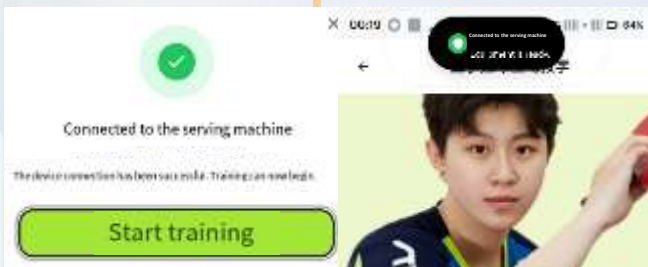


The ball-throwing machine has the ability to dynamically adjust the training plan, enabling "on-demand training" and maximizing the training effect.

Adjust the training plan dynamically

No manual operation
Automatic adjustment

Serve machine connection Placement of the Magic Mirror Camera Adjust the position



Morning:
Train the forehand smash shot

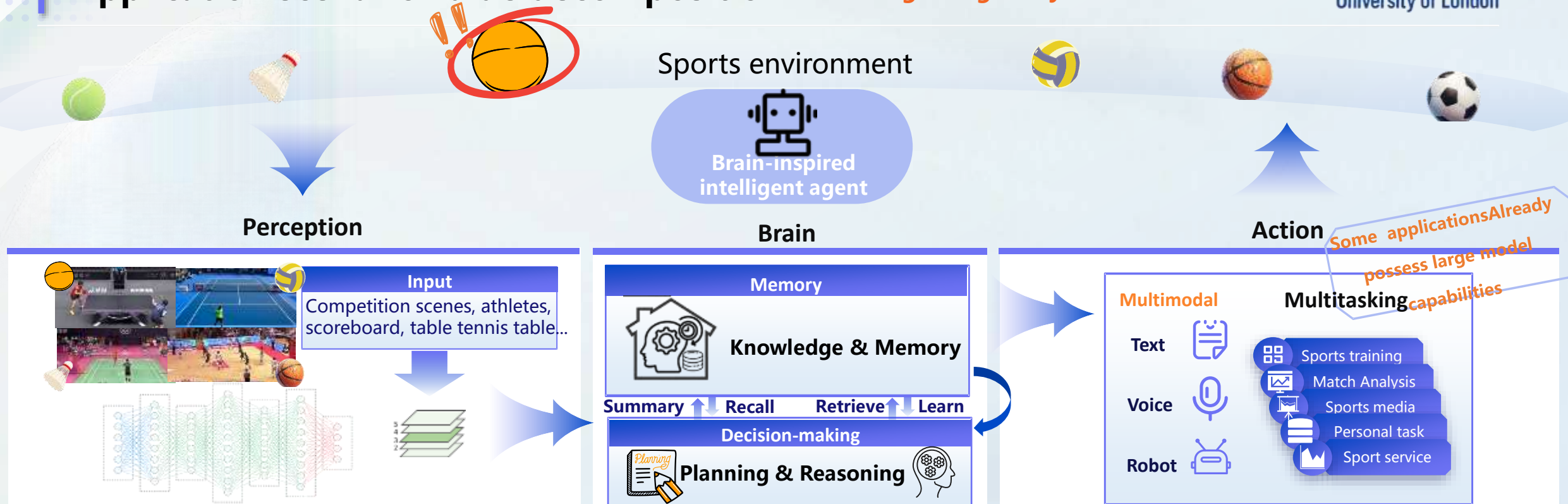
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Afternoon:
Practice backhand pull shots

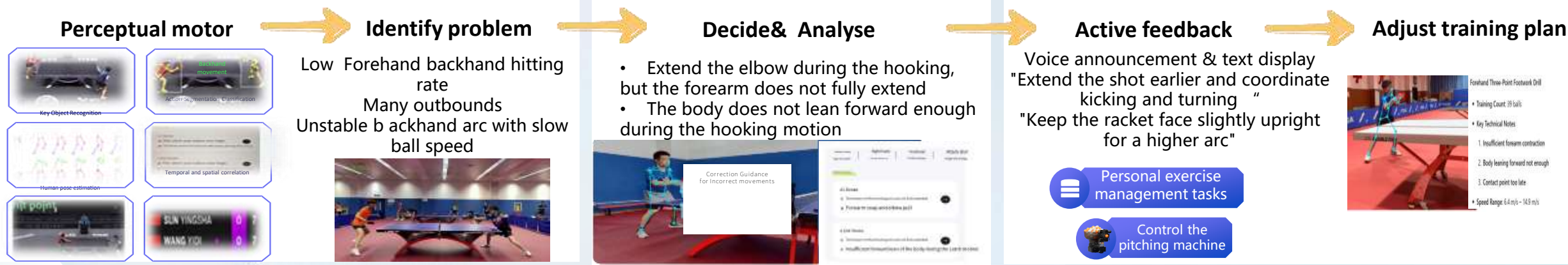


Application scenario value decomposition

Intelligent agent system



An amateur player is using the " PingPiao ZhiZhi" system to practice receiving serves.



R&d process — Personal growth, team work

2022.09

As the president of the Table Tennis Association, he worked part-time **as a table tennis coach and director of a table tennis self-media** in his spare time. He deeply experienced the pain points of the industry and found that amateur competitions were insufficient, so he conceived the idea of developing competition procedures

2022.10

I went to Beijing Sport University for **research**, communicated with teachers of our school, talked with laboratory teachers to confirm the feasibility of the product, and **conducted a questionnaire survey** among students of our school and table tennis enthusiasts from other schools

2023.03

Based on the data structures, Java and other courses **learned in class**, the team independently completed the implementation of the competition program to recruit judges, sign-in, video recording and score input

2023.09

The first application of the system in the Beijing University of Posts and Telecommunications freshman competition for team organization realized the complete event process, greatly saving the cost of event organization

2023.10

Software users **reported** that it was difficult to make progress in the competition. With the original intention of "technology for the benefit of sports", they started the project and came up with the idea of developing **AI analysis function**

Basic knowledge of the subject

- The person in charge took the lead in designing the Sai Xiaozhi mini program and discussed the specific interaction logic with the team.
- Using C and database skills from BUPT's core courses, the team independently developed mini-apps.

Technical exploration

- Report to the supervisor once every two weeks
- Group summary every four days
- A weekly student group meeting

Self-media publicity

leading official Sun Dazhuang runs a self-media for table tennis



- Cooperated with **Adam Bobrow**, a famous table tennis blogger
- Take a photo with **Geng Zhijiao**, a well-known table tennis blogger



Project leader Sun Dazhuang introduce projects to alumni foundations for investment

R&d process — Personal growth, teamwork

2023.12

The responsible person **published a paper** on the trajectory tracking algorithm of fast moving objects in table tennis based on semantic tree, which built technical barriers and laid a foundation for the subsequent commercialization of products

2024.02

Cooperate with AI direction professionals to collect **professional athletes** competition action data and train corresponding action recognition large model

2024.04

Users reported that it was difficult to **improve their own practice**, so they launched the technical and tactical analysis function to upload videos for tactical analysis and summarize their performance in the game

2024.05

AI coaching function is launched to further expand the market with **real-time video guidance**

2024.08

It is difficult to see the details of movements on the mobile screen. The magic mirror, a hardware product, has a **larger smart screen** to identify the mistakes of movements and serve beginners

Teamwork grows



The team is from the **table tennis teaching and research group of Beijing Sport University**, led by Professor Yu Yang
Discuss technical moves



World champion Liu Guozheng
Participate in the shooting of teaching courses and digital human training materials

Build a database of movements for national team athletes

Iterate the action recognition model



World champion Fang Bo
Come to the school to guide the intelligent coaches experience



Team members introduced their products to **Secretary Xu Mei** at the grand innovation exhibition

Intelligent practice mirror



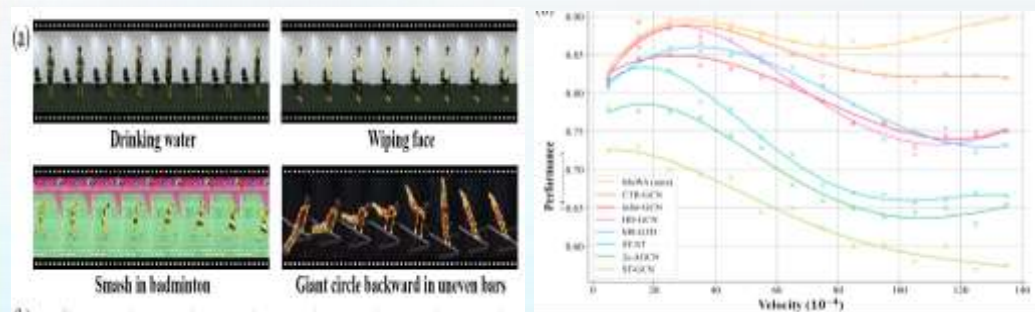
Magic mirror product features

Overview of Core Technology

Independently develop algorithms
Build a brain-like intelligent agent

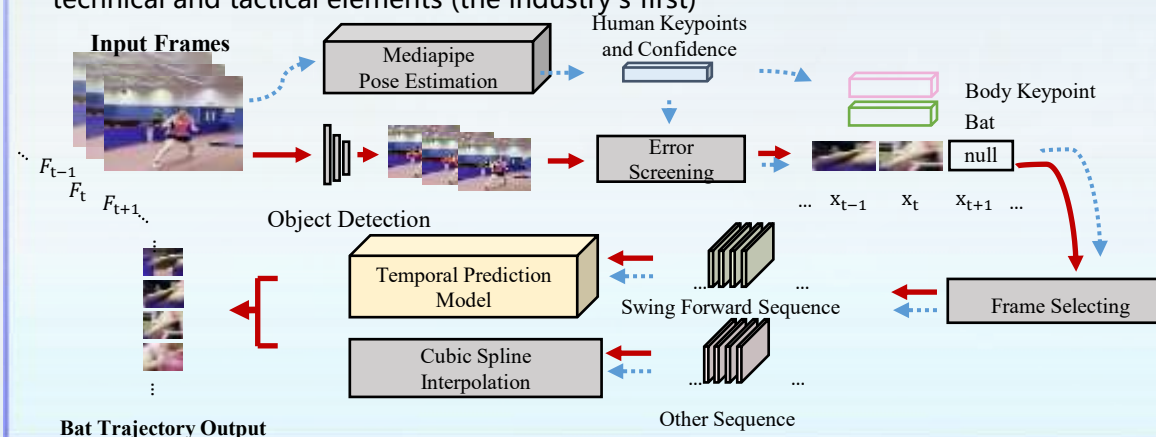
Action Recognition & Segmentation Method Based on Graph Neural Network

A speed-adaptive action classification algorithm based on Transformer (the industry's first)



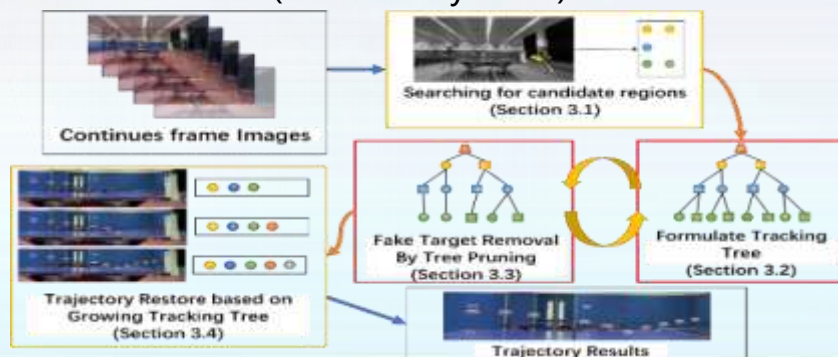
An intelligent tactical and technical analysis method covering the entire process

The complete process analysis algorithm for extracting multi-scale and multi-element technical and tactical elements (the industry's first)



A method for recognizing and reconstructing the trajectory of table tennis based on tree pruning

Tree pruning-based T-FORT high-speed sphere tracking algorithm (the industry's first)



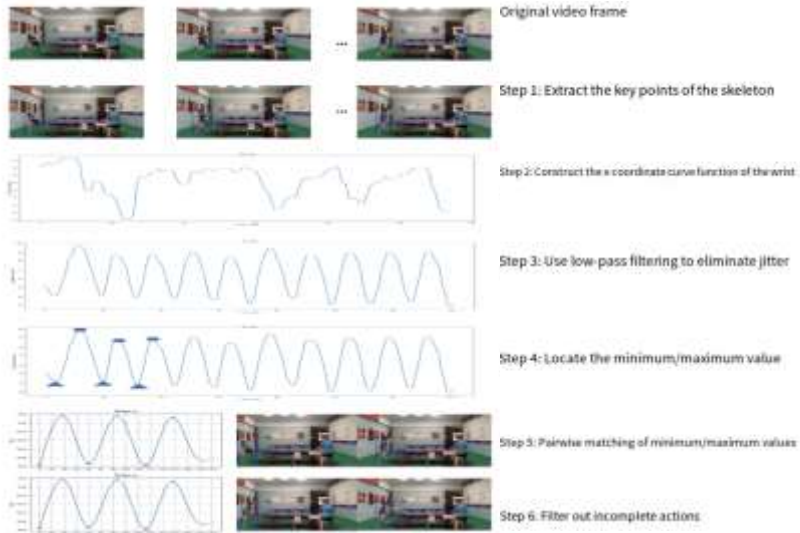
Automatic optimization algorithm for Ball Machine Full-process intelligence of Brain-like Intelligent Agent

Constructing an intelligent agent architecture resembling the human brain (industry's first) to achieve intelligent analysis



Core technology 1: Action recognition and segmentation method based on graph neural network

Swing action recognition



The innovative method developed by the team performed well in accuracy, recall rate and F1 score, with an average of 95.9%, 94.1% and 94.9% respectively, significantly better than the traditional method.

conventional method

Non-European modeling deficiency: CNN/RNN cannot model the skeletal topology relationship, and the ability of motion representation is limited

Weak feature generalization: Angle change and space-time coordination are ignored, and it is difficult to distinguish the subtle differences of hitting action

VS

Innovative approaches

Multimodal fusion modeling: ASE-GCN integrates bone and Angle features, and the accuracy of motion recognition is improved by 30%

Multi-dimensional space-time adaptation: spatial and temporal layering coordination, the classification accuracy of complex scenarios is 95%

High-order Angle quantization: Global/local Angle modeling, 40% improvement in subtle motion differentiation

1. Enter the original video of the swing
2. The two-dimensional coordinate information of 17 joints is extracted, and the two-dimensional joint coordinate data is converted into three-dimensional joint data based on the graph neural network model
3. Calculate the three-dimensional joint information such as the bending Angle of the joint
4. The calculated joint bending Angle and other joint information were compared with the ten standard action judgment rules based on the **national team members** actions
5. Determine whether the player has a wrong swing, such as overswing or high elbow, and output

Core technology 1: Action recognition and segmentation method based on graph neural network

Conventional method

- Single data dimension:** relying on a single video stream, the fine granularity of motion capture is insufficient
- Rigid and inefficient segmentation:** Fixed window threshold segmentation is difficult to adapt to complex adversarial scenarios
- Poor real-time feedback:** the analysis lags behind and lacks dynamic optimization, which restricts the training efficiency

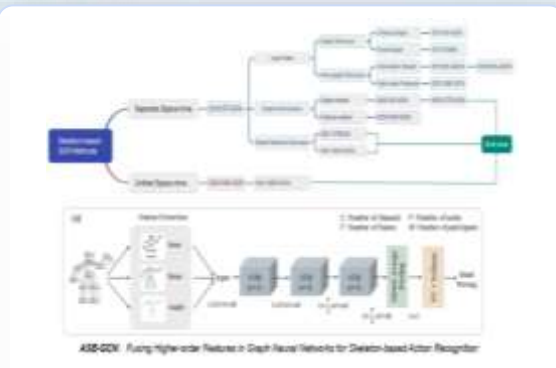
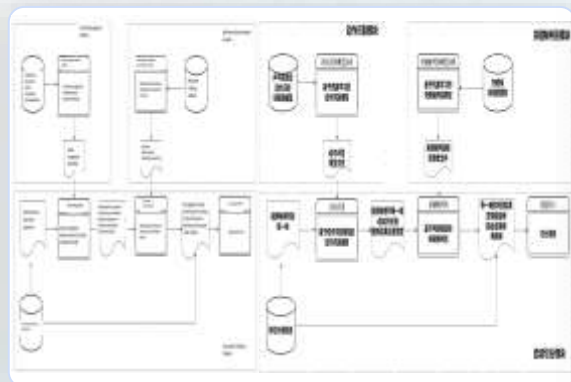
VS

Innovative approaches

- Time-space dual-stream modeling:** SlowFast heterogeneous branch batting action recognition accuracy is improved by 40%
- Robust classification optimization:** Random forest combined with smoothing and interpolation correction to deal with occlusion/sudden change of perspective
- Dynamic intelligent segmentation:** turn matching and adaptive window algorithm, adapt to complex confrontation scenarios



For the first time in the industry, video clips can be divided according to the players hitting action



Video stream auto-segmentation

Action recognition module

ARM

Based on the spatio-temporal dual-flow model, four main actions (e.g., serving, receiving preparation) of players in the game video are identified and output the actions and their confidence.

Key frame recognition module

KFDM

Based on the random forest model, all frames are classified as service start frame, loop play frame and loop end frame.

Automatic segmentation module

ASM

According to the key frame recognition results, frames are divided and combined to output effective round video clips.

The segmentation accuracy is up to 96%

Core technology 2: Table tennis recognition and trajectory reconstruction method based on tree pruning

conventional method

Weak feature capture ability: The pooling operation of FMO algorithm leads to high detection rate of small targets

High trajectory fracture rate: frame-by-frame detection relies on manual rules, and the recall rate is less than 50% in occlusion or low contrast scenarios

Low computing efficiency: The traditional method has high complexity, limited real-time and scalability, and is difficult to adapt to multi-scenario requirements

VS

Innovative approaches

Multi-frame differential anti-fog: dynamic differential + adaptive threshold, accurately capture high-speed targets, and improve the accuracy of fuzzy scenes by 43%

Tree trajectory modeling and pruning: semantic tree structure + motion consistency score, fracture rate reduced by 50%, complexity $O(N)$

Graph network enhanced generalization: GCN replaces backtracking classification with attention fusion mechanism, and the recall rate of trajectory in complex scenarios is 92%



Compared with the accuracy of mainstream FMO algorithm

Up 43%

01

Input frame set

02

Search for the target ball in the candidate area

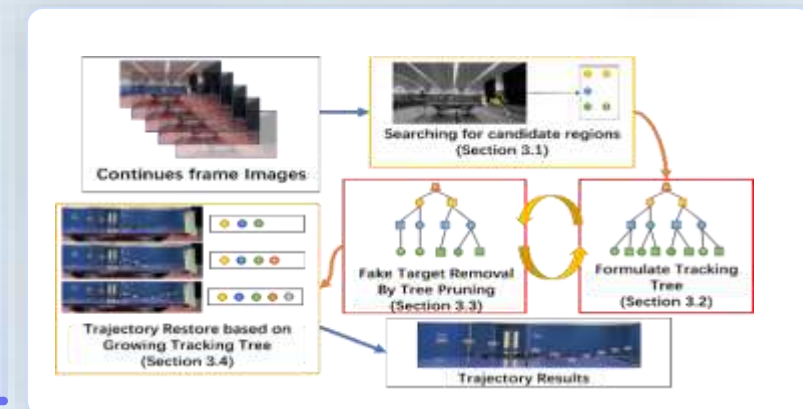


03

Generate trajectory results

04

Trajectory recovery based on the spanning tree



Core technology 2: Table tennis recognition and trajectory reconstruction method based on tree pruning

conventional method

High hardware cost: traditional eagle eye requires professional sensors and high-speed cameras

Complex deployment: The installation and calibration process is cumbersome, and it is only available for professional venues

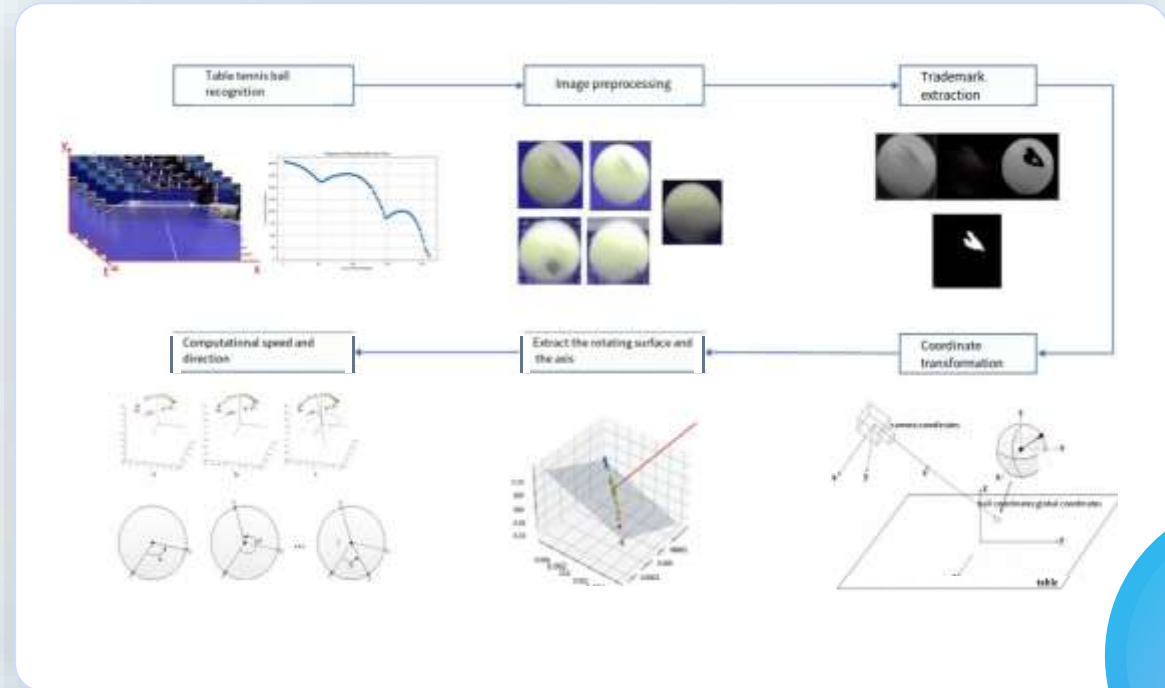
Scenario limitations: It is difficult to meet the needs of daily training and public scenarios

VS

Innovative approaches

Single-eye camera detection: ordinary mobile phone can realize rotation detection and hitting point analysis

Low cost and high adaptability: it can be adapted to mobile phones, tablets and other devices, and is available to all



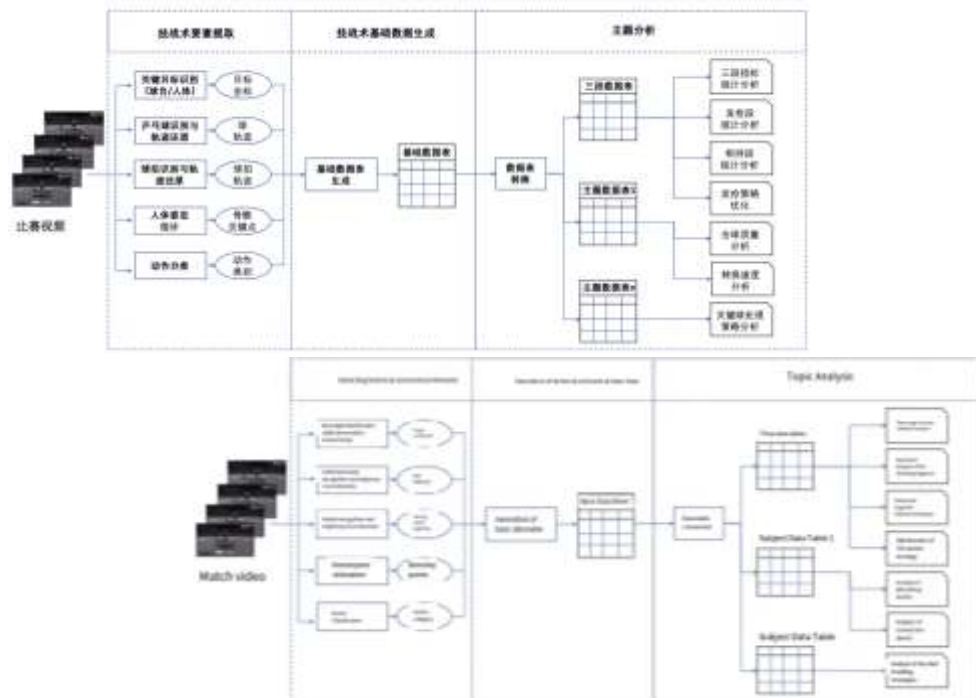
Rotational
detection
Schematic
diagram of
principle



Schematic diagram of ball strike detection effect

Core technology 3: Intelligent technical and tactical analysis method for the whole process

The team annotated the training videos provided by the national team for AI model training



Forehand Three-Point Footwork Drill

- Training Count: 39 balls
- Key Technical Notes:
 1. Insufficient forearm contraction
 2. Body leaning forward not enough
 3. Contact point too late
- Speed Range: 6.4 m/s – 34.9 m/s

Make AI become a professional table tennis player

- 1. Enter the original video of the swing
- 2. Use the recognition and analysis modules to extract technical and tactical elements.
- 3. Generate technical tactical basic data table
- 4. Analyze hitting and conversion speeds during rush and hold phases, then provide suggestions.

Conventional method

The analysis dimension is single: only the stage scoring rate is counted, ignoring the dynamic details such as the number of rallies and ball speed

Low efficiency of manual labor: relying on coaches to annotate frame by frame, which is subjective and difficult to support high-frequency training optimization

VS

Innovative approaches

Full-process analysis: covering the whole stage from serving to holding, and quantifying the static limitations of the "three-stage method"

Multimodal fusion: skeleton + ball trajectory + 23 types of motion recognition, accurate analysis of tactical chain logic

Tactical knowledge base construction: Based on the case of Sun Yingsha, a reusable game strategy model is generated.

Core technology 4: Engine automatic optimization algorithm

Multi-brand interface compatibility



At present, mainstream brands of wireless devices on the market can be connected to Ping Xiaozhi

hardware layer

Multi-source sensor collaboration

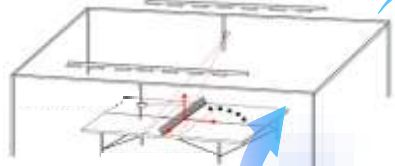
Sensor type	technical parameter	functional objective
High speed binocular vision	120Hz frame rate + 2 megapixels (3000fps scalable)	Ball trajectory/landing point/rotation speed capture
9-DoF IMU	Acceleration + gyroscope + magnetometer triaxial fusion	Kinetic parameters of hitting action were collected
Synchronous trigger module	Fiber optic transmission + μ s level timestamp alignment	Solve the problem of multi-source data space-time alignment

How can AI wake up traditional devices?

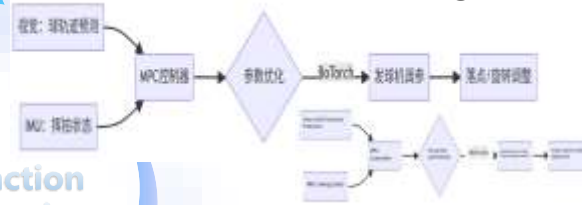
— The enabling logic of Ping Xiao Zhi

AI algorithm closed loop

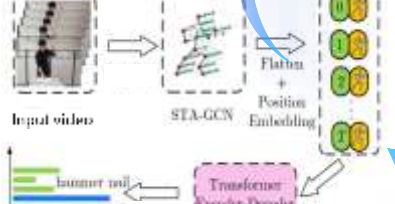
Data collection



Real-time adjustment of serving parameters



Intelligent interaction
Technological iteration



Analyze the weaknesses of users

Generate training plan



Software layer

Mechanical duplication

Technological iteration

Intelligent interaction

Intelligent voice control

With the intelligent voice interaction system, users can adjust training parameters in real time by voice command without manual operation

Technical summary

The integration of multiple technologies empowers the improvement of precision

Action recognition and segmentation method based on graph neural networks

2023.12 propose

Action recognition accuracy, recall rate and F1

Average: **95.9%、94.1% and 94.9%**

2024.05 upgrade

Action recognition **Increase by 30%**

Complex Scene Classification

Accuracy rate: 95%

Differences in subtle movements

Increase by 40%

Input the original video



A method for recognizing and reconstructing the trajectory of table tennis based on tree pruning

2023.10 propose

Compared with the mainstream FMO algorithm, **Accuracy increase by 43%**

2024.02 upgrade

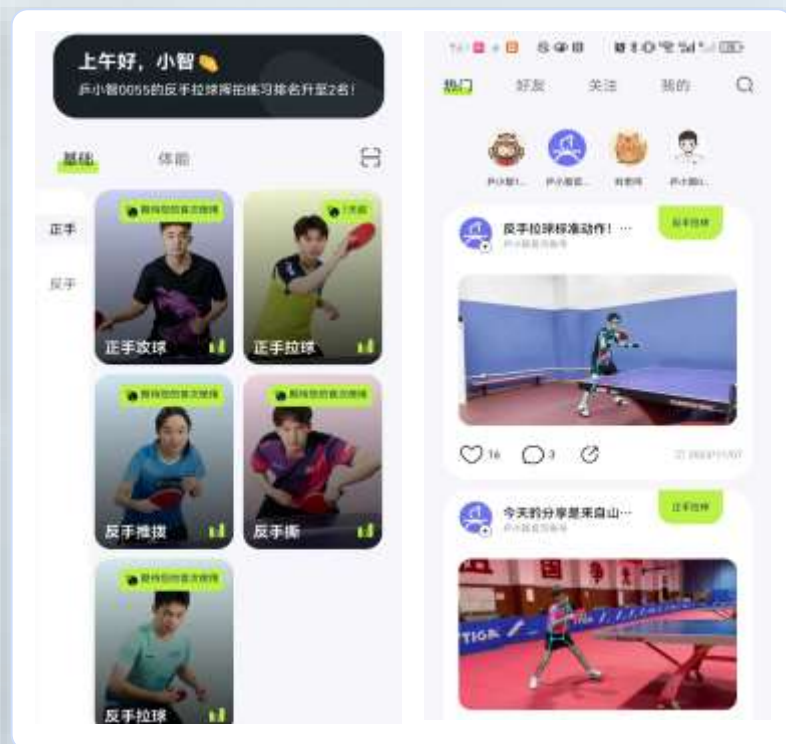
Tracking accuracy **increase by 43%**

Breakage rate **decrease by 50%**

Complex scene trajectory **recall rate 92%**

Technology implementation

Products targeted at amateur enthusiasts



Summarize the performance of the competition

Unearth tactical issues

Based on the training of national team athletes' movements

Self-developed intelligent analysis model for table tennis movements



AI Table Tennis Coach Mirror

Intelligent practice mirror

Advanced Course

Movement correction

Teaching and practice



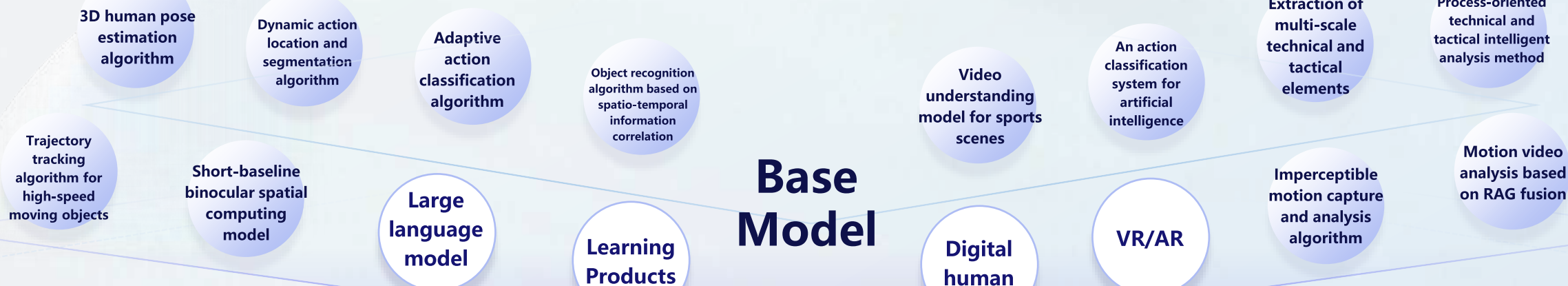
Features of the Magic Mirror product

- High-definition camera capture action
- Enjoy comfortable training on a large screen
- Ultra-precise modeling of human body sensing

Team achievements

Based on the most advanced technologies, the team has successfully developed intelligent moving agents and achieved remarkable results

Base Model



Team members, first inventor: National Invention Patent (Publication Number: CN116246202A)

One has been authorized, four are under application, and four computer software copyright registration certificates are held

The industry ranks first in peer review for sports movement classification, movement correction, ball tracking, and racket tracking indicators

Published one high-level journal paper

A Semantic Web-Based Approach for Bat Trajectory Reconstruction With Human Keypoint Information

Authors: Wu, Beijing Sport University, China
Yuh-Zheng, Beijing University of Posts and Telecommunications, China
Jian-Lin, Beijing University of Posts and Telecommunications, China
Yang-Yu, Beijing Sport University, China

ABSTRACT

Reconstructing the trajectory of a bat from a table tennis tennis technique and conducting statistical analysis frame is challenging due to changing shapes cause leading to ambiguity. This paper develops a novel method for bat detection in each frame, followed by fitting stages, the authors use a temporal prediction model interpolation to reconstruct a complete bat trajectory and performance are evaluated on our video data proposed method outperforms traditional method screening algorithm improves precision score to 22.3% higher than YOLOv5 and also 1.4% higher than



在第二十六届中国机器人及人工智能大赛全国总决赛中,表现优异,脱颖而出,荣获 人工智能创新赛

一等奖

团队成员: 姚江宇, 孙大伟, 张智宇
指导老师: 刘军
特发此证, 以资鼓励。

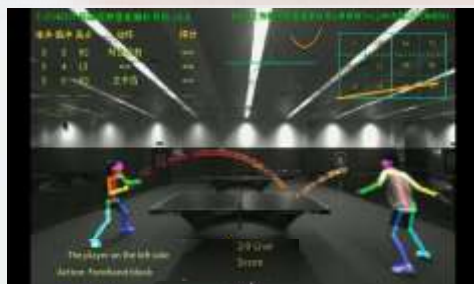
证书编号: CHAC2024010010

中国机器人及人工智能大赛组委会
二〇二四年八月

Cooperation achievements of the national team

Help the national team achieve breakthroughs

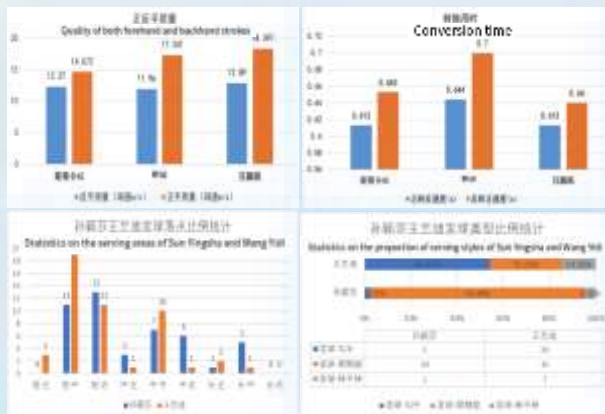
The team has already reached a deep cooperation agreement with the Chinese national team, jointly implementing the 2024 major scientific and technological project. **They collect data from the national team to train the model.** They provide intelligent and data-driven training support to the national team, significantly improving training efficiency and competitive level, and consolidating its absolute leading position on the international stage.



Collect data of the national team players from various world-class competitions and input it into the model. Communicate the research results **with the head coach of the world's top-ranked player, Sun Yingsha.**

Coach Liu Guozheng of the men's **national team** came to the school to guide the development of the system

World champion Fang Bo collaborates and communicates with our team.



5"

This partly depends on the equipment, but the main factor lies in practice.

Did you make any significant technical adjustments to the backhand during the training?

There are still some changes.

Increased proactive efforts



孙颖莎王艺迪发球落点比例统计

1. 双方均使用发球落点比例统计软件，使用该软件进行数据统计。
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日期：2024.1.1

Signed an agreement with the Science Research Institute of the National Sports Administration

Project outcome - Widely praised globally

China's first intelligent sports system that integrates independent algorithms and hardware and is exported overseas

National Collegiate Table Tennis Association



The Polish national team and former player Wang Zengyi tried out the Pingqi Smart System and gave it a high rating.



Austrian club uses AI function to assist training



Spanish national youth team members experience the paddle-swinging practice function of



Organizing international table tennis exchange activities between China and other countries



Overseas user data



26.5w+
views

3.4w+
Experience user
application

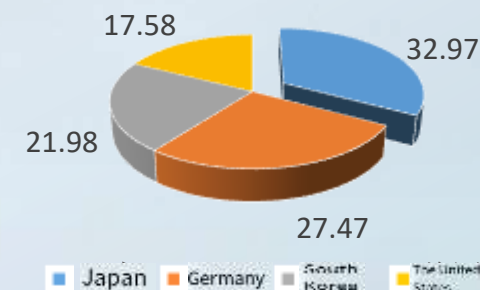
58712
Participants of the
competition

3500+
Conducted the
competition altogether

8700+
Tactical analysis for
users

Millions +
Expected users

The proportion of overseas users



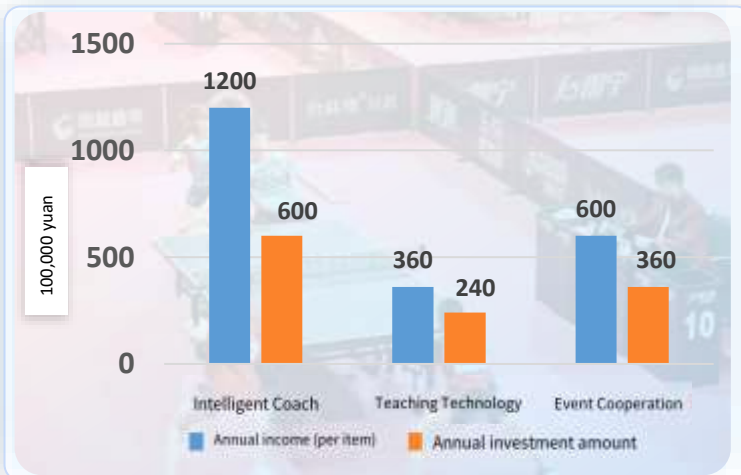
Development planning

2026: Deepen product development. 2027: Achieve full commercialization

2025-2026年

The world's first "AI + intelligent serving machine" hardware product is launched.

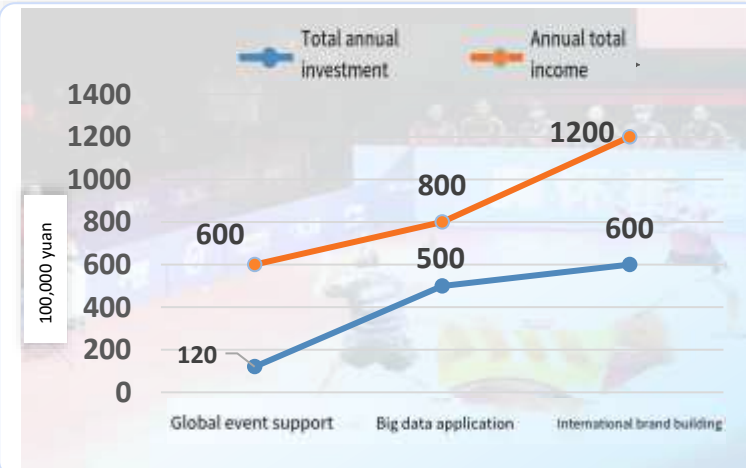
- The championship-level coaching algorithm matrix powers the hardware, and iteratively enables the application of the Pingxi Xiaozhi intelligent agent.
- Completed the joint test of 10 national professional athletes, completed market verification, and completed the angel round financing.
- Launch global social media operations, expand cooperation with multiple renowned events, and conduct comprehensive marketing promotion.
- Launch the first-generation mass production to quickly capture the market, and provide services for over 10,000 enthusiasts.



2026-2027年

Establish a global hardware hegemony system

- Log in to the top 10 sports equipment channels in North America/Europe.
- By applying and driving the iteration of infrastructure, the technical indicators of the intelligent movement system have ranked first in the world.
- Complete the signing of benchmark partner agreements for the intelligent agent platform, and enhance the international influence of the brand in this field.
- Become the "official intelligent training partner" for the WTT World Cup and ITTF Challenge tournaments



2027-2028年

Take the lead in the global smart sports equipment market

- Signing contracts with the top 50 global table tennis clubs and 32 national teams
- Global event empowerment: Supported by an intelligent serving machine system, over 8,000 professional events in the industry have been held.
- Introduce VR virtual reality technology, release the self-evolving training system, and engage in real-time battles with global enthusiasts
- Participate in and sponsor major international events and activities, achieve global market coverage, and consolidate the leading position of the brand

The penetration rate of AI+ sports domain intelligent agent capabilities has exceeded 50%, achieving a significant lead in the industry and generating an annual revenue of 200 million yuan.



A company is about to be established, focusing on small ball sports with barriers

Project valuation: 50 million yuan

The alumni sponsorship provided an initial fund of **1 million yuan**

Self-funding 30% + Venture capital 70%; **3** investment institutions have already conducted evaluations.

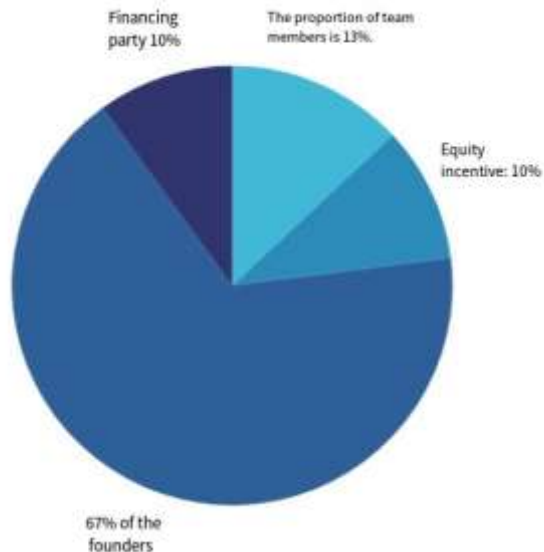
By highlighting the project's key features and expected value-added benefits, we seek investors for investment.



The system's net profit margin exceeds **30%**.

The team holds **80%** of the overall equity.

Equity incentives account for **10%**.



Revenue growth

2023 year 0 yuan →
2025 year **140000** yuan

Cost control

Research and development as well as hardware account for **60%**, and after scaling up, the marginal cost will decrease by **15% to 20%**.



**Huishui Hongyuan (Hangzhou)
Technology Partnership Enterprise**

Business model

Integrating the upstream and downstream of the industrial chain, AI reconfigures the industrial value chain

Technical partner

Provide data collection and algorithm development support

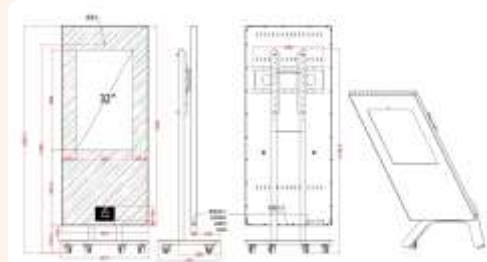
Cloud service provider

Provide servers and cluster computing power

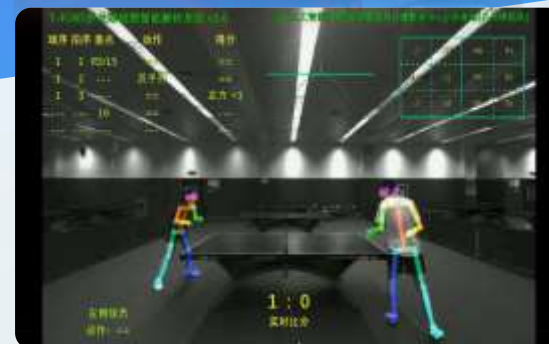
Electronic goods supplier

Provide basic components and sensors

Implementation of hardware manufacturing and cloud service guarantee technologies



Upstream: Resource Supply and Technical Support



B2B2C

Data collaboration loop Business users

Applied in primary and secondary schools, colleges and training institutions, to enhance the quality of table tennis teaching

Applied to clubs and event organizers, it optimizes the entire process of event organization.

Consumer users

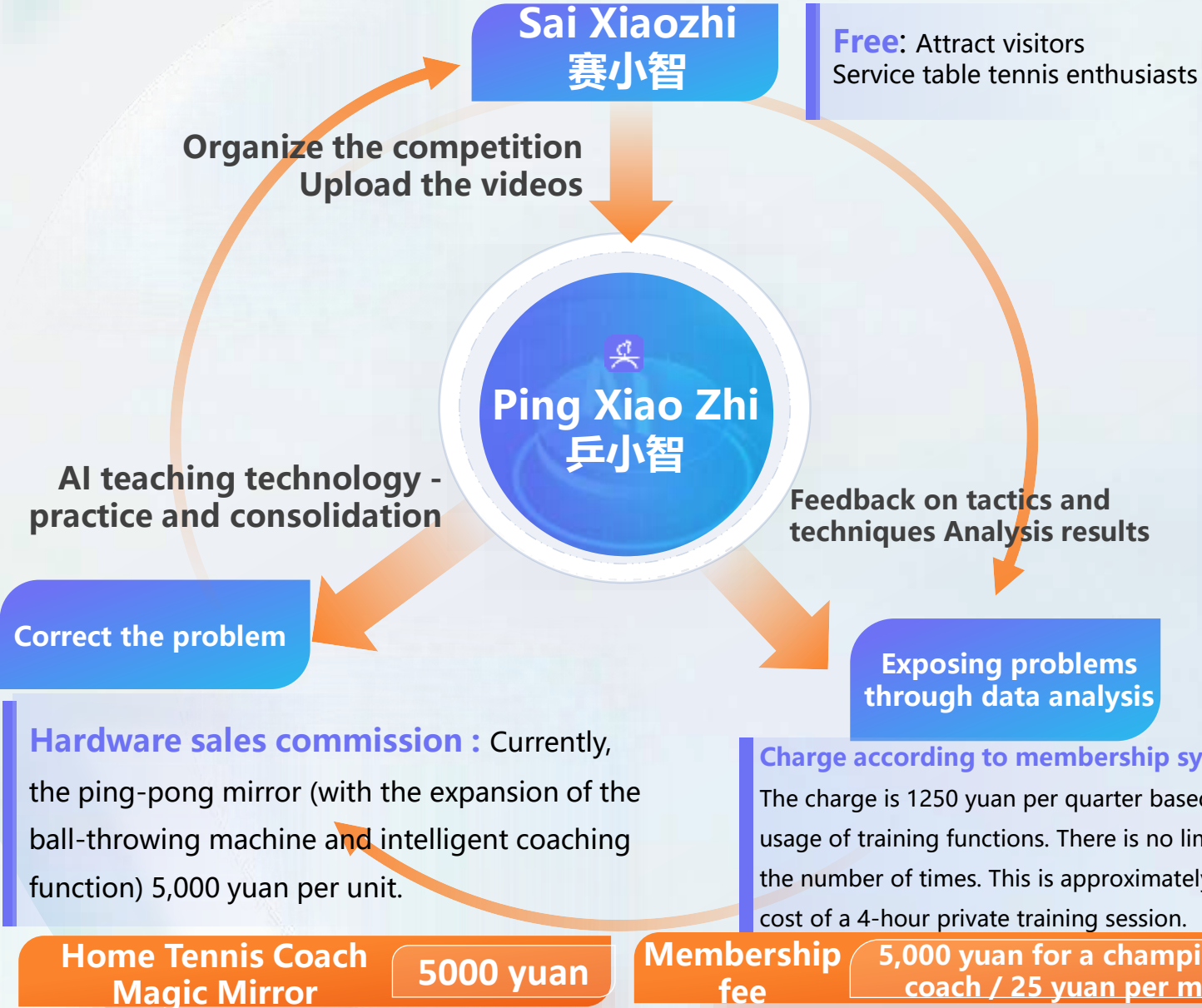
Provide professional technical guidance for amateur table tennis players
Provide detailed technical and tactical analysis for professional athletes



Downstream: User-oriented technological inclusiveness

Business model

Integrated revenue from learning, training and competition
Outlook on new technologies of intelligent agents



Future operation outlook: Provide intelligent agents for game commentary

During the live broadcast of the game, through AI's automatic editing, automatic statistics, and analysis of movements and tactics, the entertainment value of watching the game can be enhanced.

Data statistics and intelligent question answering

Based on the self-developed intelligent model for understanding competition videos, automatically understand and analyze the broadcast content.



Digital human intelligent commentary

Support the addition of intelligent digital human commentary for video competition resources.



Charge by usage

Charge based on the usage of interface calls

1,000,000 users * 3,000 k/year * 0.01 yuan

30 million yuan/year

Charge on breakdown

Divide the revenue according to the income of the partner users

2 million users * 10 yuan/year in revenue sharing

20 million yuan /year

Charge mode

Diversified marketing, with the goal of achieving an annual revenue of 200 million yuan within three years

Autonomous Intelligent Agent Application

Charge mode	Charge by subscription	Charge by usage	Hardware charging sales
Details of Charges	There are two charging methods: regular membership and champion coach membership Regular members pay 50 yuan per month, and champion coaches pay 2,500 yuan per year	Charges are based on the usage of training functions It costs 2 yuan per use, with 100 uses per year, totaling 200 yuan, which is approximately the cost of one hour of personal training	Currently, we are selling the Table Tennis Magic Mirror (with an extended serving machine linked to an intelligent coach).8,000 yuan per unit
Estimated revenue	50 yuan per month * 100,000 users + 2,500 yuan per year for 5,000 users 50 million yuan per year	2 yuan per time * 100 times * 400,000 users 80 million yuan per year	8,000 yuan per unit for 4,000 customers 32 million yuan per year

Specific charging model

Ordinary member **50** yuan per month

- only includes AI interactive courses

Champion Coach Member **2500**

- Champion Coach AI interactive courses and training plans
- 1 online one-on-one video guidance per week
Monthly offline centralized face-to-face guidance X1
- Every month, there will be a lucky draw for autographs of national players

Artificial intelligence technology

X

Home fitness equipment

Table tennis coach Magic mirror



Price: **8000** yuan per unit



One year of championship coach membership will be presented as a gift, along with in-class assistance training plans and assessment functions for schools and clubs

Revenue forecast: It is estimated that within three years, the annual revenue will reach 200 million yuan.

System comparison

Ping Xiao Zhi is far ahead.

	 Ping Xiao Zhi Support for connection	 Traditional ball machine	 Alpspace Technology Intelligent Coaching System	 Keep series of fitness and health apps
Technical framework	The three-layer architecture of brain-inspired intelligent agents: perception (multimodal data) - brain (dynamic knowledge base) - execution (software and hardware across multiple terminals)	Basic Serving Platform, supporting external connection	AI + Machine Vision (Standard Action Model))	× Lack of technical analysis ability in table tennis
Core function	National-level technical analysis - AI system "Ping Xiaozhi" and "Magic Mirror" for real-time error correction - Personalized transfer training	 - No action analysis function - Requires an external system for training feedback	 - The action model is simple - There is no built-in training plan - The supported actions are few, and there is no rotation-related function	- Generalized motion data statistics - Standardized courses
Data capability	Global Sensing: Professional end (millimeter-level motion capture) / General public end (Magic Mirror vision + APP linkage)	 Not equipped with AI analysis capabilities	 High deployment cost	Wearable device integration (heart rate / step frequency)
Application scenarios	- Competitive (National Team Technical Optimization) - All citizens (Family/Community/School) - Industrial Standardization	 Provide the foundation for the AI training system in terms of playing environment	 The hardware cost is 50,000 yuan. It is solely targeted at the B sector and currently there is no mature product available.	 Lack of national team data and technical support
Core advantage	National team data foundation National team-level technical recognition - Compatible with hardware, strong implementation capability	Compatible with mainstream training support systems	Fun Interaction	Billions of users + Ecological integration

Product illustration



Compatible with traditional ball machines

Production mode and sales channels

Integrate online and offline resources and expand the market in all aspects

Light-asset R&D mode

- **Software development:** Based on the technical agile iteration of the Beijing University of Posts and Telecommunications laboratory (2-6 weeks per version)
- **Algorithm training:** National team professional movement data + provincial athlete data set
- **Hardware production:** Devices such as the Magic Mirror adopt OEM outsourcing and integrate self-developed software systems

B-end institution channels (core traffic entry points)

- ▶ **Sports stadium/club partnership**
 - **Promote the Ping-Pong Magic Mirror hardware**, leveraging the exclusive AI motion analysis technology to enhance the teaching efficiency of coaches and the training quality of students.
 - We have signed contracts with **7 chain ball halls** including Beijing Boming Table Tennis and Huaqi Table Tennis.
- ▶ **Infiltration of the school / Table Tennis Association**
 - Through the connection of the Table Tennis Association of Beijing University of Posts and Telecommunications to the "Beijing University Table Tennis Alliance", the Magic Mirror is deployed as an intelligent teaching tool to **enhance the quality and efficiency of grassroots teaching**. Provide **AI-assisted teaching and scientific level assessment** for physical education courses and school teams.
 - Jointly build a "AI+ Table Tennis" pilot project **with the Table Tennis Association of Haidian District** and establish a government-subsidized procurement system.

Direct sales channels for the C end (the main battlefield for paid conversions)

- ▶ **A closed-loop system of study, practice and competition**
 - Users organize competitions through **Sai Xiaozhi**, upload videos, and expose problems through data analysis of techniques and tactics. By using the **Ping Pong Magic Mirror** and **Ping Xiaozhi** AI teaching technology to study courses, practice and consolidate, and correct problems, it effectively helps amateur enthusiasts improve.
- ▶ **Penetration of online and offline scenarios**
 - Online, through KOL videos on Douyin and Kuaishou for **traffic diversion and AI battle reports** on official accounts for viral growth, we have established a path of "**viewing → experiencing → paying**" with professional content and zero-threshold experience.
 - Offline, we have set **up scan-to-use experience zones** in cooperative tennis venues and **offer weekly cards for on-site registration** at university table tennis competitions, comprehensively guiding users to immerse themselves in the closed loop of learning, practice and competition.



Marketing promotion

Online media promotion matrix

Sai Xiaozhi made it to the front page of China Education Daily



CCTV publicity report



CCTV10 has invited team members and mentors to participate in the "Innovation in Progress" program.



Technical support for the top ten balls of the National Games

Ping Xiao Zhi provides AI analysis technical support for the **top ten ball performances for the National Games**.



Cooperation with
broadcasters



Provide game commentary intelligent agent capabilities for broadcasters

Cooperation between
table tennis
manufacturers



Self-media
cooperation

Equipment
review

Teaching
cooperation

Self-media
promotion

Reach cooperation with
millions of big V users



Application Case

Popularize the advanced sports technologies of the national team to the general public through AI

The table tennis magic mirror assists physical education classes in primary, secondary and tertiary schools



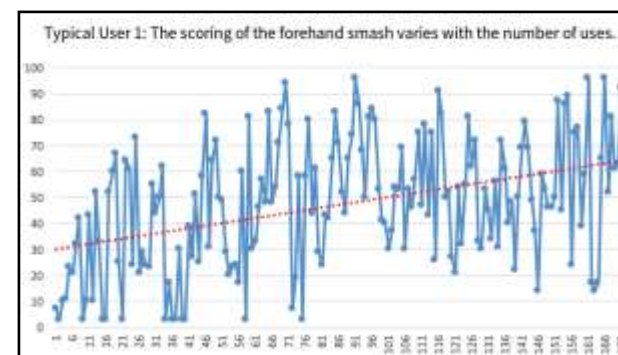
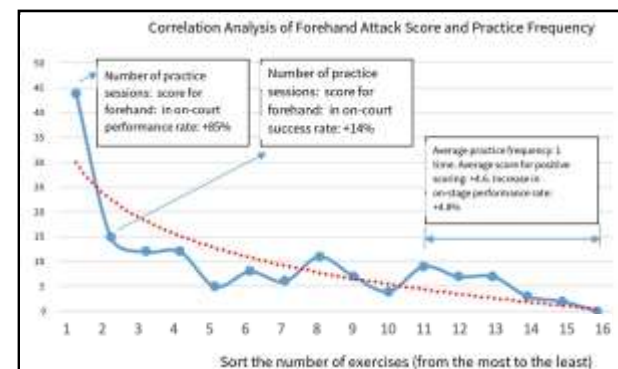
Provide smart classroom support for students of Changshun High School in Guizhou Province and Sanfan High School in Beijing. Students learn table tennis techniques in class through **Pingxiaozhi**, and the AI coach corrects their movement mistakes in real time



It is applied in physical education classes at Beijing University of Posts and Telecommunications, Beijing Sport University and Beijing Normal University, using the table tennis magic mirror coach to **capture and analyze** users' table tennis movements. Students learn to adjust their swing postures according to **the voice guidance in the magic mirror**



The students' classroom performance has significantly improved based on the national team data training standard movement model



116 students, 1368 swings of the racket

International market

The wind of table tennis is blowing across the world

The perfect integration of
locality and globality
one AI coach for every culture

Fully adapt to the habits of foreign users



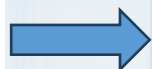
The core of overseas amateur training is the combination of social interaction and leisure, which is different from the domestic focus on competition. However, the flexibility of our products can be fully adapted to record sports anytime and anywhere, facilitating personal improvement

Due to the professionalization level of table tennis overseas, professional coaches are in greater shortage than in China. Our products meet the high-level guidance needs of foreign athletes for training, as well as the professional one-on-one analysis that is more difficult to achieve abroad

Compatible with international brands
and in line with international standards

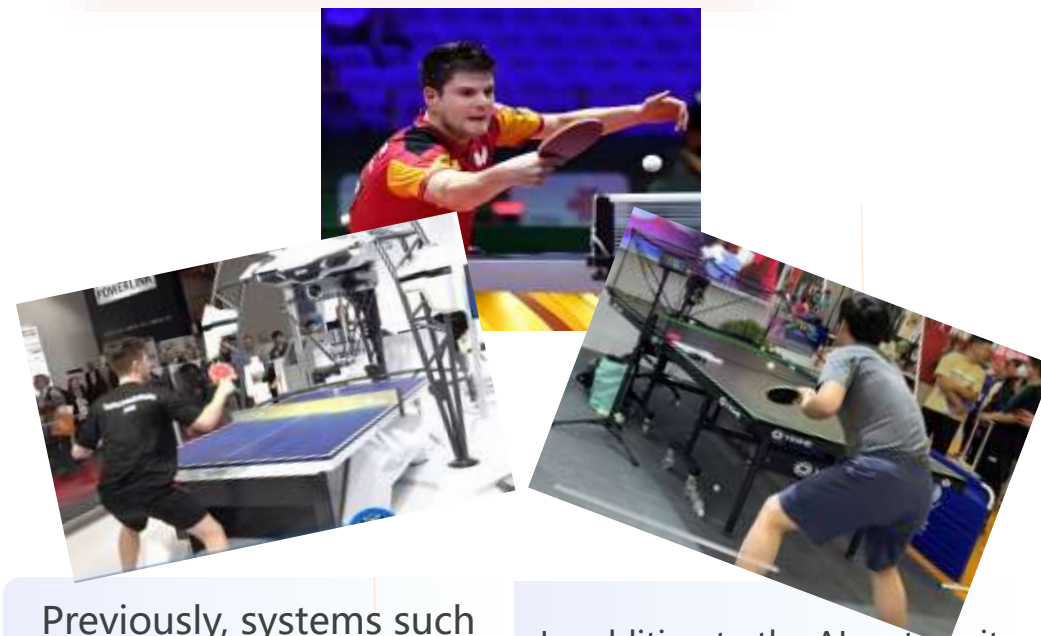


PONGBOT



Covering major overseas ball machine brands, it offers voice guidance in over 20 languages including English and Spanish

Overseas promotion has taken root



Previously, systems such as **Pingxiaozi** have been promoted in dozens of countries overseas and have over 50,000 participating users

In addition to the AI system, it has also held over 3,000 competitions and numerous table tennis exchange activities, receiving widespread acclaim

Team members

The multi-disciplinary integrated innovation team provides solid technical support and comprehensive expansion both at home and abroad



Weiye Li

Team Leader

Intelligent Science and Technology

Undergraduate student of Queen Mary University of London. Received the **National First-Class Scholarship** and the honorary title of "Outstanding Student". Once awarded the title of National College **Student Mathematics competition** First Prize, Third Prize in the National College Students' English Competition, Second Prize at the **Beijing Provincial level** in ICT. A member of the college basketball team.



Tangrui Zhang

Project Content Planning

Information and Computing Science

Undergraduate student of Queen Mary University of London. A die-hard sports enthusiast, he won the **third place in the badminton competition** of BUPT and the **second place** in the badminton competition of Bielefeld University of Applied Sciences in Germany. He has **served as a judge** for the table tennis league of BUPT many times. Participated in the **technical development** of the voiceprint recognition project of the Young Goose Program and is a multilingual coder.



Leyan Li

Product Manager

Intelligent Science and Technology

Undergraduate student of Queen Mary University of London. Passionate about the field of artificial intelligence, with a **solid foundation in programming**, **skilled in back-end development**, having in-depth research on the competition systems of badminton and table tennis, and proficient in using programming languages such as C, Python, and Java. Was once a **member of the debate team**.



Huishun Hu

App Development

Internet of Things Engineering

Undergraduate student of Queen Mary University of London. Has **won the Bronze Award in the China International College Students Innovation Competition**, the title of **Outstanding Student Leader**, the **Second-Class Scholarship at the university level**, and the Third Prize in the National College Students **Mathematics Competition**. Proficient in using programming languages such as C and Java



Dazhuang Sun

Overall Technology Development

E-commerce and Law

Undergraduate student of Queen Mary University of London. A hardcore table tennis enthusiast. So far, **four papers** related to table tennis movement recognition and correction, table tennis trajectory tracking, occlusion detection, and competition points have been **published**, and **one national invention** patent has been authorized. During his tenure as the **president of the Table Tennis Association** of BUPT, the association was awarded the title of "Top Five and Top Ten" for three consecutive years.



Yaodan Yan

Copywriting and Marketing

Telecommunication Engineering and Management

Undergraduate student of Queen Mary University of London. A **member of the IET Society**, proficient in multiple programming languages such as C, C#, and python, and passionate about the field of **artificial intelligence**. Be responsible for most of the subsequent improvement work in the team and **analyze users and related data**.



Xuan Bai

Front end and HCI Design

Telecommunication Engineering and Management

Undergraduate student of Queen Mary University of London. She has won the third prize in the National College Students' **English Competition** and is proficient in programming languages such as C and JAVA. Currently, he is a **member of the Chuyan Project** at Beijing University of Posts and Telecommunications.



Chengrui Li

Development of AI algorithms

Sports Engineering

Undergraduate student of Vienna University of Technology. **100,000 followers of table tennis self-media** worldwide. The men's singles champion of the **Riha Open** and the **captain of the table tennis team** of Vienna University of Technology. **Top three** in the Sports Engineering major.



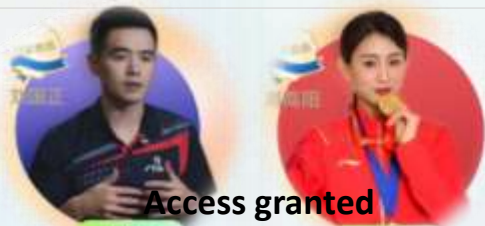
Yiming He

Development of AI algorithms

Mathematics

Students of the University of Texas at Texas. Was awarded the title of **Honorary Student** by the college. He possesses profound expertise in the tournament structures of badminton and table tennis, complemented by strong programming fundamentals and proven proficiency in back-end development.

University mentor World champions guarantee the professionalism of table tennis techniques



Have joined the coaching team of
Ping Xiaozhi Champions

Guozheng Liu

Champion Coach

Associate Professor of
Tsinghua University
International-level athlete

Table tennis world champion, head
coach Wang Chuqin, and former head
coach of the men's national table
tennis team.

The digital coach image of the
champion spokesperson has been
launched



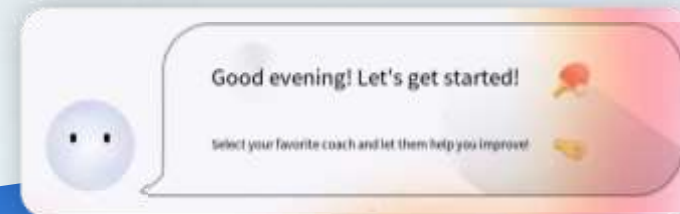
Gaoyang Liu

Gold Medal Coach

Professor of Shandong
University of Finance
Economics International-level
athlete

Champion of the Youth Olympic Games

Provide motion data and professional
experience Provide testing and trial
conditions for intelligent technical and
tactical analysis software, and
participate in its optimization and
improvement





Algorithm
Guidance

Jun Liu

Professor and table tennis head coach at Beijing University of Posts and Telecommunications; also Vice-President, Table Tennis Branch, Beijing University Students' Sports Association. Published papers in SCI, EI and leading Chinese journals; led students to awards in the China "Internet+" Innovation & Entrepreneurship Competition; PI for key projects of the China Higher Education Society; chief editor of table-tennis textbooks. As head coach, secured 10 world, 80 national and 105 provincial/ministerial medals, bringing honor to the university and the capital.



Table tennis
guidance

Hao Zhang

Professor and table tennis coach at Beijing University of Posts and Telecommunications; also vice president of the university sports association's table tennis branch. Published in SCI, EI and top Chinese journals, led students to awards in the national "Internet+" competition, headed key China Higher Education Society projects, and co-authored table-tennis textbooks. My teams have won 10 world, 80 national and 105 provincial medals, bringing honor to Beijing and the university.



Entrepreneurship
guidance

Kelian Chen

Vice Dean of the School of Artificial Intelligence, Beijing University of Posts and Telecommunications; Deputy Director of the Digital Economy Innovation and Entrepreneurship Research Center, Beijing University of Posts and Telecommunications. An expert in multimodal artificial intelligence, he has presided over multiple projects funded by the National Natural Science Foundation of China and guided undergraduate and postgraduate students to win over ten national and provincial-level innovation and entrepreneurship awards as well as disciplinary awards

Advisory team

Professional athletes contribute to the scientific nature of data



There are many current and retired key players of the national team

About to join Ping Xiaozhi
Champion Coach Course



Fei Xue
Current athletes of
the national team



Shuo Yang
Retired athletes of
the national team



Xu Du **Sports guidance**

Edited the textbook "Research on
Table Tennis Teaching Theory and
Training Methods"



Dandan Xiao

Technical director of
the national team



Na Ja

International Gold Medal referee
The highest level of table
tennis referee work



Yang Yu

Former coach of the men's
national table tennis team
Head of the Table Tennis
Teaching and Research Group
of Beijing Sport University
The leading figure in academic
research on table tennis in universities



Luming Gong

Vice Chairman of the Chinese
Basketball Association

Meritorious coach of the
Chinese national basketball
team



Kai Guo **Business expert**

International College of Beijing University of
Posts and Telecommunications

Expert judge of the China International
"Internet Plus" College Students Innovation
and Entrepreneurship Competition

Has won the first prize of Beijing Higher
Education Teaching Achievement Award

With a heart for the country and the people, we are deeply engaged in the field of intelligent agents and intelligent sports, shouldering the mission of empowering the health of all citizens with technology

Cultivate a deep sense of patriotism and love for the country



Team members integrate their personal development into national needs and translate the ideal of "I will contribute to a strong country" into actions of "serving the country through science and technology", fully demonstrating the mission and responsibility of the youth team of Beijing University of Posts and Telecommunications in serving the national strategy in the field of artificial intelligence.

Cultivate the spirit of innovation



The project has received **key cultivation and support** from the school's physical education classroom plan and the National College Students' Innovation and Entrepreneurship Competition, and has always been characterized by an innovative spirit, entrepreneurial awareness, and the ability to innovate and start businesses. Based on solid professional skills, apply what has been learned **in the field of agents**.

Training on the integration of science and technology innovation, industry and innovation



The team closely combined scientific research achievements with engineering practice, completing the transformation and implementation from scientific research prototypes to engineering products with application value. With the support of the school's "Top-notch Innovative Talent Cultivation" initiative, focusing on the pain points such as the **low level of intelligent training and the lack of auxiliary systems**, the core technologies were closely combined with actual needs, successfully breaking through the technical bottlenecks of **real-time and precise human-computer interaction**.

Cross-disciplinary and cross-border innovation



The team breaks down the knowledge barriers between specialties such as artificial intelligence and mechanical control, giving **rise to innovative methods of multimodal integration and intelligent perception**. Facing complex engineering problems, it enhances technical depth, broadens thinking, and strengthens engineering practice in the intersection of multiple disciplines. The research and promotion of the project have provided a solid support for the cultivation of **"compound, innovative and engineering"** talents, forming a virtuous cycle of "scientific research feeding back to teaching and project-driven growth".

Focus on the common growth of individuals and teams, build a capability growth system, create future-oriented creativity

From the classroom to practice, apply what has been learned in class, innovate and transform achievements, and comprehensively advance abilities

Case 1



computer programming

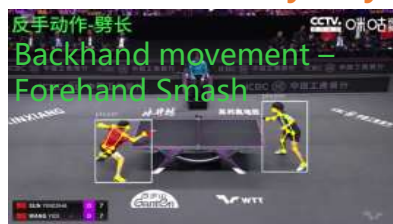
Case 2



Deep learning training model

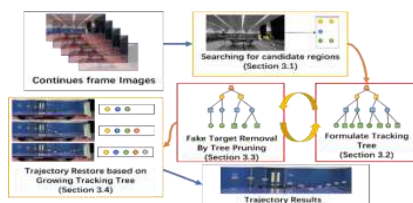
Technical and tactical analysis system

Technology transfer



1 patent
4 software copyrights

Table tennis trajectory recognition model



One high-level journal paper

A Semantic Web-Based Approach for Bat Trajectory Reconstruction With Human Keypoint Information

Accumulation of
hardcore technology

Advanced integration
of resources

Expansion of
Business Thinking

Has won over 20 national-level competition awards including the first prize in the China Robot and Artificial Intelligence Competition, over 30 provincial and ministerial-level competition awards, and the National Scholarship

Team members participated in 8 scientific research internships, including those at the National Disaster Recovery Center, the National Key Laboratory of Optical Communication, and the Low Earth Orbit Satellite Communication Laboratory

A number of scientific research achievements have been produced in the field of artificial intelligence, with a total of 6 papers, 1 patent and 4 software Copyrights published

Team members have won the Silver Award for volunteer service at the China Open and other awards, with a total of 976 hours of volunteer service

The team grows and succeeds



A Semantic Web-Based Approach for Bat Trajectory Reconstruction With Human Keypoint Information

Zachuan Liu, Beijing Sport University, China



The "platform - venue - fund" trinity support provides a solid guarantee for the project's advancement

Interdisciplinary support

The Artificial Intelligence Laboratory of **Beijing University of Posts and Telecommunications** has a deep cooperation with the Table Tennis Teaching and Research Group of **Beijing Sport University**. People from Beijing University of Posts and Telecommunications provide **artificial intelligence technology**, while those from Beijing Sport University offer **professional knowledge of table tennis**.

From the Grand Innovation Exhibition to actual training scenarios, creating a benchmark model of education through industry-university-research collaboration.

Members of the AI major and the sports major **complement each other's strengths, promote each other, and improve together**.



Facilitate personal growth

Participate in the public relations of the project

The Table Tennis teaching and Research group of Beijing Sport University and the students of the Artificial Intelligence Laboratory of BUPT have carried out **interdisciplinary project applications**, expanding the development of artificial intelligence in the field of sports and providing strong technological assistance for the scientific training of sports projects.

Support for key platforms

Relying on the national-level innovation and entrepreneurship base of the university, it has **received resource support from the key laboratories of the International College and the School of Artificial Intelligence**.

The **supercomputing center** provides thousand-core-level computing power support and the data resources of **the national-level sports members of the school team** are fully open throughout the process.

With the strong support of **the school's physical education department**, the project was **awarded the highest priority for incubation** in the "Innovation and Entrepreneurship WIN Base".



With the support of the school's "comprehensive training model", the project team has achieved a coordinated growth of dual improvement in scientific research capabilities and engineering practice, gradually evolving into a young scientific and technological force with an interdisciplinary perspective and independent innovation capabilities.



WELCOME TO THE AGENTIC SPORT ERA